Determination of Normal Skin Elasticity by Using Real-time Shear Wave Elastography

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Objectives—To define the reference ranges of normal skin elasticity measurements associated with shear wave elastography (SWE) in healthy volunteers and analyze the factors that may affect SWE.

Methods—Mean skin thickness and elastic modulus values from 90 healthy volunteers were evaluated with B-mode ultrasonography and SWE in the right fingers and forearms, anterior chest, and abdominal walls. Reference ranges of normal skin elasticity were calculated by using lower and upper limits at the 2.5th and 97.5th percentiles. To investigate the effects of potential factors (site, sex, age, body mass index, and skin thickness) on skin elasticity measurements, a 1-way analysis of variance, the Student *t* test, and the Pearson correlation test were performed.

Results—Skin elasticity was significantly different at different sites ($P < .05$). Mean elastic modulus values were 30.3 kPa for the finger, 14.8 kPa for the forearm, 17.8 kPa for the chest wall, and 9.5 kPa for the abdominal wall, and reference ranges of normal skin elasticity were 12.1 to 48.4kPa for the finger, 3.5 to 26.0 kPa for the forearm, 6.6 to 28.9 kPa for the chest wall, and 3.5 to 15.5 kPa for the abdominal wall. Our study revealed that men had higher skin elasticity measurements than women ($P < .05$), and they were more elevated in participants aged 20 to 50 years than in the other groups at the finger ($P < .05$). The body mass index and skin thickness had a negligible impact on skin elasticity measurements ($P > .05$).

Conclusions—This study revealed that the site, sex, and age should be taken into account when determining the reference ranges of normal skin elasticity by skin elasticity measurements.

Key Words—dermatology; elasticity; normal; shear wave elastography; skin; ultrasonography

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Abbreviations
BMI, body mass index; ROI, region of interest; SWE, shear wave elastography; US, ultrasonography

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Skin thickening and tightness are characteristic manifestations of systemic sclerosis due to excessive dermal deposition of collagen, and the degree of skin involvement correlates with the severity of internal organ manifestations, a poor prognosis, and increased disability.[1,2] Successful clinical management of systemic sclerosis requires correct diagnosis and evaluation of the degree of skin involvement. In particular, noninvasive, reproducible methods to evaluate the degree of skin involvement are crucial for repeated monitoring of the disease and therapeutic responses.[3,4]

Current noninvasive methods for assessing skin involvement in systemic sclerosis include the modified Rodnan skin score,[5,6] magnetic resonance imaging,[7] high-frequency B-mode ultrasonography

(US),[2,8–10] a number of other mechanical methods,[3,4,11–13] and ultrasonic elastography. Among these methods, the modified Rodnan skin score, which is based on palpation, has been favored by most clinicians; indeed, it requires no special equipment and is accessible, noninvasive, and cost-effective.[4] However, its role in the evaluation of scleroderma remains controversial because of its operator dependency, which leads to high intraobserver and interobserver variability. Moreover, this method cannot differentiate between skin thickness and tightness.[2,5,6,11] Other mechanical tools, such as the torsional device, elastometer, suction cup, and durometer, have mostly been used as research tools or to commercially assess the effects of emollients and are not widely applied in routine clinical practice.[3,4,11–13] Measurements with the suction cup, elastometer, or torsional device may take more than half an hour for the entire body, and the skin of body parts with an uneven surface (eg, fingers), is not appropriate. Durometer measurements vary considerably when the pressure or speed of the indenter onto the specimen is not constant.[3,4] Magnetic resonance imaging and US are additional tools for evaluating the degree of skin involvement. Magnetic resonance imaging may have some limitations; for example, it requires a long acquisition time and is not cost-effective. Meanwhile, high-frequency US is so cost-effective, noninvasive, and portable that clinicians use this technique to estimate the extent of skin thickness while roughly evaluating the stage of systemic sclerosis, such as edema, fibrosis, or atrophy.[2,8–10] Nevertheless, B-mode US cannot measure skin elasticity.

Newer methods, such as strain elastography, shear wave elastography (SWE), and acoustic radiation force impulse imaging, allow the assessment of skin elastic properties.[14–21] Disadvantages of strain elastography include operator dependency, low reproducibility, and its relative qualitative evaluation.[16] Among the above techniques, SWE and acoustic radiation force impulse imaging, being the most recent methods, possess several advantages, such as real-time B-mode (2-dimensional) imaging, lack of operator dependency, and quantitative measurements. Therefore, they may be useful to differentiate clinically affected skin from healthy skin and monitor the degree of skin fibrosis or therapeutic responses in patients with systemic sclerosis.[16,18,20,21] Moreover, Lee et al[18] showed that SWE provided better differentiation and less variability compared with acoustic radiation force impulse imaging in distinguishing sclerotic lesions of morphea from the normal dermis. Our previous study confirmed that SWE has high intraobserver and interobserver repeatability for normal skin elasticity (intraclass and interclass correlation coefficients of 0.885 and 0.908, respectively).[21]

Thus, to accurately interpret results in patients with systemic sclerosis, a reference range of normal skin elasticity assessed by SWE in healthy individuals is required. In addition, the factors that may affect skin elasticity must be evaluated, as differences in the site, sex, age, body mass index (BMI), and skin thickness in healthy individuals should be recorded. To the best of our knowledge, no study has tackled this issue. Thus, the purpose of this study was to determine the normal reference values of skin elasticity for SWE parameters, evaluating the associated effects of age, BMI, skin thickness, sex, and site in healthy volunteers.

Materials and Methods

Volunteers

This study was approved by the Ethics Committee of West China Hospital of Sichuan University (approval No. ChiCTR-DCD-15006851) and performed in accordance with the Declaration of Helsinki of 2013. Between September 2015 and December 2015, 90 healthy volunteers were evaluated by B-mode US and SWE. Volunteers were recruited according to the following exclusion criteria: pregnancy in women; scar at the measurement site; history of skin, rheumatoid, immune, metabolic, or endocrine diseases; and history of radiotherapy or chemotherapy. No additional tests were performed on the volunteers. The following parameters were recorded at the time of the study: age, sex, weight, height, and BMI.

Ultrasonography

The participants were instructed not to perform any form of exercise 2 hours before the examination, with complete rest for 5 minutes before the procedure. We examined 4 sites according to our previous study,[21] numbered as follows: site 1, right middle finger (dorsum of the middle phalanx); site 2, right forearm (dorsal aspect, 10 cm proximal to the ulnar styloid); site 3, anterior chest (between the sternal angle and sternal notch); and site 4, anterior abdomen (10 cm distal to the sternum). Sites 1 and 2 were examined with the volunteers putting their hands and forearms on the examination bed, with palms down in a naturally flexed state without applying any strength. Sites 3 and 4 were examined with the volunteers in the supine position, with both hands on the waist and shoulders relaxed. Each volunteer was required to hold his or her breath for a moment to minimize the breathing motion.

Ultrasonography was performed by a sonographer with experience in musculoskeletal US using an Aixplorer US system (SuperSonic Imagine, Aix-en-Provence, France), with an SL 15–4 multifrequency linear transducer operating at 7.5–15 MHz, which has a resolution of 0.3 mm in the B-mode and 1 mm in the SWE mode. The superficial musculoskeletal setting was selected, with the "standard" default mode of the instrument; the SWE elasticity range was adjusted to 100 kPa for the best image quality according to our previous study.[21] The SWE settings were configured as follows: 50% opacity and auto time-gain compensation. The tip of the transducer was covered with several millimeters of US gel and placed perpendicularly to the skin smoothly in a longitudinal section without applying any pressure on the skin. All volunteers had a normal skin structure, defined as a high epidermal echo, medium dermal echo, and low subcutaneous echo, with no tumors, pseudotumors, or inflammatory lesions, and with distinct and parallel interfaces between them according to previous reports.[21–23] B-Mode US was used to measure the skin thickness at all 4 sites, including the epidermis and dermis. Skin thickness was measured 3 times at the same location, averaged, and expressed in millimeters.

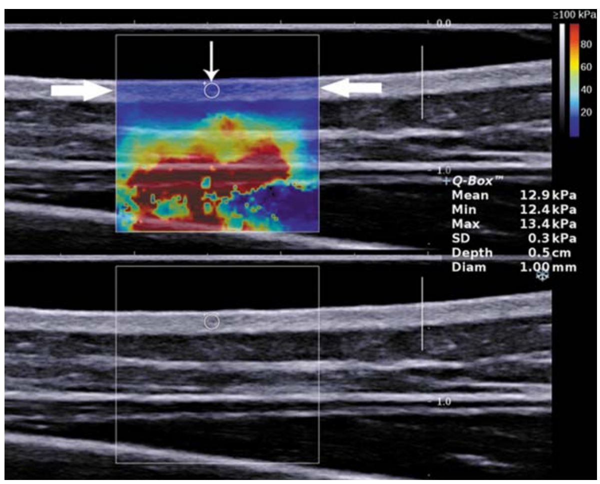
After shifting to the SWE mode, regions of interest (ROIs) for recording SWE could not be placed in a different site for the B-mode examination. Although the elastic modulus value displayed on the color image is not affected by the ROI size, standardization of the ROI size and location was performed as follows: the superficial and deep ROI margins were manipulated to include the area from the gel to the subcutaneous tissue. At each site, the transducer was held for several seconds during image acquisition until the SWE in the ROI was stable in color, and the image was then saved. A round Q-box adjusted to the skin thickness was placed in the skin on the color-coding SWE image (1 Q-box measurement per image), and the system automatically calculated the mean, maximum, minimum, and standard deviation of elastic modulus values for the Q-box area in kilopascals (Figure 1). The mean value was selected as the representative value for each image. The transducer was maintained at the same position for 3 measurements (1 Q-box measurement per image) at each skin site; the averaged elastic modulus values were recorded in kilopascals.

Statistical Analyses

Statistical analyses were performed with SPSS version 19.0 software (IBM Corporation, Armonk, NY). The Kolmogorov-Smirnov test and Q-Q plot were used to assess the normality of continuous variables. To evaluate the effects of possible factors, including site, sex, age, BMI, and skin thickness, on skin elasticity measurements, a 1-way analysis of variance for multiple-group comparisons (site and age groups), the Student *t* test for a binary variable (sex), and the Pearson correlation test for continuous variables (BMI and skin thickness) were used.

The lower and upper limits of the reference range were estimated at the 2.5th and 97.5th percentiles, respectively, of the distribution of the mean skin elastic modulus values in the volunteers. When the reference values followed a normal distribution, the lower and upper limits of the reference range were mean elastic modulus value – $1.96 \times$ SD and mean elastic modulus value + $1.96 \times$ SD, respectively, where SD is the

Figure 1. Representative SWE (top panel) and B-mode US (bottom panel) images of skin in the forearm obtained by a linear transducer operating at 7.5 to 15 MHz. The SWE range was adjusted to 100 kPa for the best image quality (top right in the SWE image). On the SWE image, a color map of relative skin elasticity is shown in the SWE box (thick arrow). The mean, minimum, maximum, and standard deviation of skin elasticity in the Q-box (thin arrow) are presented on the right side of the image. The mean elastic modulus value of the skin was 12.9 kPa.



standard deviation of the mean elastic modulus values. Two-sided $P < .05$ was considered statistically significant.

Results

Study Population

The main characteristics of the 90 volunteers are summarized in Table 1. In this study, the 90 participants with normal skin comprised 45 women and 45 men. The mean age ± SD was 35 ± 25 years (range, 3–89 years). The mean body weight was 52.01 ± 18.60 kg (range, 14–87 kg); the mean body height was 1.53 ± 0.21 m (range, 0.91–1.82 m); and the mean BMI was 21.37 ± 4.53 kg/m^2 (range, 12.5–30.22 kg/m^2). Mean skin thickness values differed for the finger, forearm, anterior chest wall, and anterior abdominal wall (1.07 ± 0.14, 1.41 ± 0.22, 1.80 ± 0.45, and 1.94 ± 0.46 mm, respectively); the differences were statistically significant ($P < .001$) with the exception of the chest wall–abdominal wall pair ($P = .193$).

Influence of the Site

Mean skin elastic modulus values were significantly different in the distinct sites, at 30.3, 14.8, 17.8, and 9.5 kPa for the finger, forearm, chest wall, and abdominal wall, respectively ($P < .05$; Table 2 and Figure 2). The reference ranges of normal skin elasticity were 12.1 to 48.4 kPa for the finger, 3.5 to 26.0 kPa for the forearm, 6.6 to 28.9 kPa for the chest wall, and 3.5 to 15.5 kPa for the abdominal wall (Table 2).

Influence of Sex

Mean skin elasticity characteristics by sex are shown in Table 3 and Figure 3. Mean skin elastic modulus values for male participants were significantly higher than those for female participants at all sites, except for the finger. The reference ranges of normal skin elasticity were 14.1 to 46.9 kPa for the finger, 5.1 to 28.5 kPa for the forearm, 6.9 to 23.9 kPa for the chest wall, and 3.8 to 13.9 kPa for the abdominal wall in female participants and 16.1 to 52.8, 6.5 to 28.3, 6.6 to 30.3, and 5.2 to 19.6 kPa, respectively, in male participants.

Table 1. Characteristics of the 90 Healthy Volunteers

Characteristic	Value	P^a
Volunteers	90	
Female	45 (50)	
Male	45 (50)	
Age, y		
All volunteers	35.32 ± 24.53 (3–89)	.788
Female	36.02 ± 24.88 (4–89)	
Male	34.62 ± 24.44 (3–79)	
3–19	30 (33.3)	
20– 50	30 (33.3)	
>50	30 (33.3)	
BMI, kg/m^2		
All volunteers	21.37 ± 4.53	.499
Female	21.05 ± 4.78	
Male	21.70 ± 4.30	
Skin thickness, mm		
Right finger	1.07 ± 0.14 (0.70–1.75)	
Right forearm	1.41 ± 0.22 (0.95–2.10)	
Chest wall	1.80 ± 0.45 (1.00–2.60)	
Abdominal wall	1.94 ± 0.46 (1.00–2.70)	

Data are presented as mean ± SD (range) and number (percent) where applicable.
[a]Student *t* test.

Table 2. Skin Elastic Modulus Values for the Healthy Volunteers by Site

	Elastic Modulus, kPa			
Site	Mean ± SD	Range	95% CI	Reference Range
Right finger	30.3 ± 9.3	14.0–53.0	28.3–32.2	12.1–48.4
Right forearm	14.8 ± 5.8	5.0–28.0	13.6–16.0	3.5–26.0
Chest wall	17.8 ± 5.7	6.5–30.5	16.6–19.0	6.6–28.9
Abdominal wall	9.5 ± 3.1	3.8–19.9	8.8–10.1	3.5–15.5

The difference is statistically significant between any 2 groups ($P<.05$) by a 1-way analysis of variance and Student-Newman-Keuls analysis. CI indicates confidence interval.

Figure 2. Skin elastic modulus values assessed by SWE in healthy volunteers at different sites.

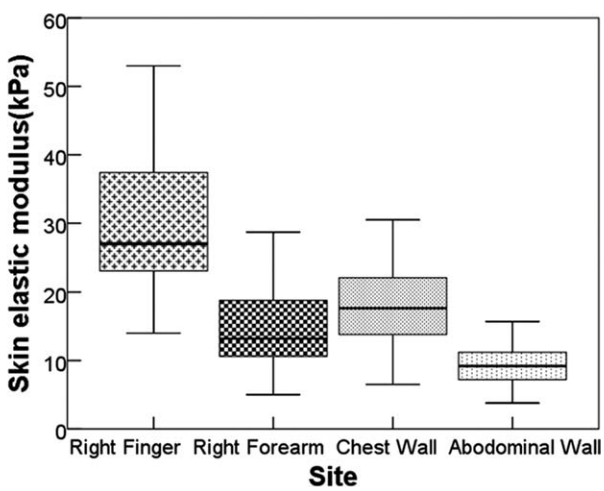
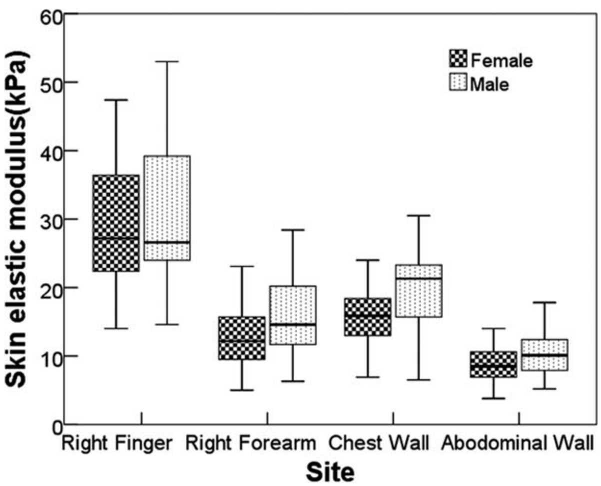


Table 3. Skin Elastic Modulus Values for the Healthy Volunteers by Sex at Different Sites

		Elastic Modulus, kPa		
Site	**Sex**	**Mean ± SD**	**Range**	**95% CI**
Finger	Male	31.8 ± 10.0	15.4–53.0	28.8–34.8
	Female	28.7 ± 8.3	14.0–47.4	26.2–31.2
Forearm[a]	Male	16.1 ± 5.5	6.3–28.4	14.5–17.8
	Female	13.4 ± 5.8	5.0–28.7	11.7–15.1
Chest wall[b]	Male	19.9 ± 6.1	6.5–30.5	18.0–21.7
	Female	15.7 ± 4.3	6.9–24.0	14.4–16.9
Abdominal wall[b]	Male	10.4 ± 3.3	5.2–19.9	9.4–11.4
	Female	8.6 ± 2.5	3.8–14.0	7.8–9.3

CI indicates confidence interval.
[a]$P < .05$.
[b]$P < .01$.

Figure 3. Skin elastic modulus values assessed by SWE in healthy volunteers by sex at different sites.



Influence of Age

Mean skin elasticity characteristics by age are shown in Figure 4. At the finger, mean skin elastic modulus values were significantly different in various age groups, at 25.2, 36.1, and 29.6 kPa in volunteers aged 3 to 19, 20 to 50, and >50 years, respectively. At the forearm, statistically significant differences were only observed between volunteers older than 50 years and the remaining volunteer cohorts. Mean elastic modulus values were insignificantly different in various age groups for the chest and abdominal walls.

Influence of BMI

The BMI was not significantly correlated with the skin elastic modulus for the finger, forearm, anterior chest wall, and anterior abdominal wall (finger, $r = 0.189$; $P = .074$; forearm, $r = -0.110$; $P = .301$; chest wall, $r = 0.019$; $P = .863$; and abdominal wall, $r = 0.021$; $P = .841$).

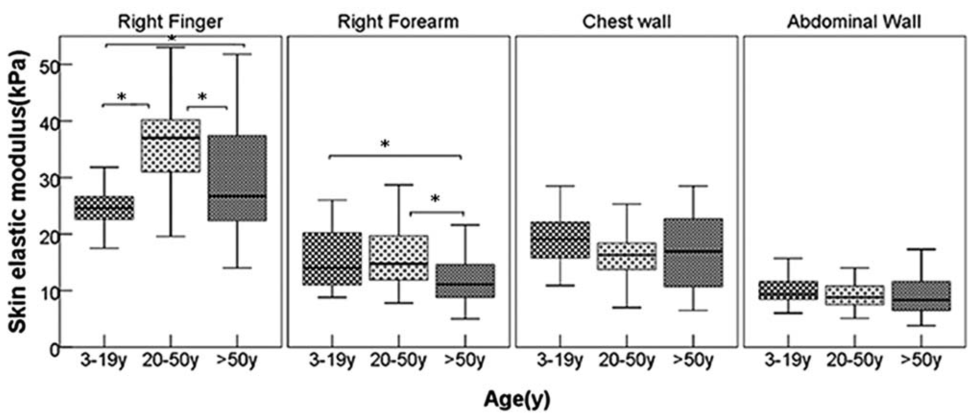
Influence of Skin Thickness

Skin thickness was not significantly correlated with the elastic modulus for the finger, forearm, anterior chest wall, and anterior abdominal wall (finger, $r = -0.020$; $P = .849$; forearm, $r = 0.006$; $P = .955$; chest wall, $r = 0.004$; $P = .972$; and abdominal wall, $r = -0.017$; $P = .875$; Figure 5).

Discussion

In this study, we measured normal skin elastic modulus values by SWE in 90 healthy volunteers. The main results were that the site, sex, and age had significant impacts on skin elasticity. Normal skin elasticity differed

Figure 4. Skin elastic modulus values assessed by SWE in healthy volunteers by age at different sites.



significantly according to the site of measurement (finger > chest wall > forearm > abdominal wall). Reference ranges for normal skin elasticity were 12.1 to 48.4 kPa for the finger, 3.5 to 26.0 kPa for the forearm, 6.6 to 28.9 kPa for the chest wall, and 3.5 to 15.5 kPa for the abdominal wall, averaging 30.3, 14.8, 17.8, and 9.5 kPa for the finger, forearm, chest wall, and abdominal wall, respectively. Normal skin elasticity values in male participants were significantly higher than those of female participants, except for the finger. At the finger, normal skin elasticity significantly differed among the age groups (20–50 > >50 > 3–19 years). At the forearm, normal skin elasticity was significantly lower in participants older than 50 years compared with the values of other cohorts. The BMI and skin thickness of the participants did not affect the SWE results.

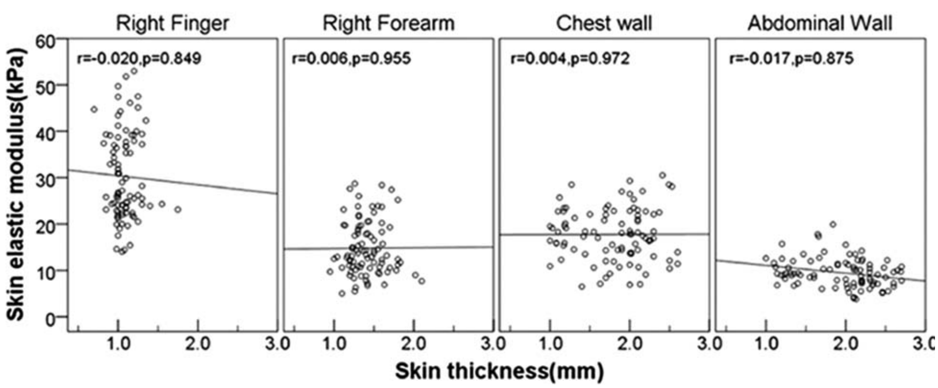
This study had several unique features. First, to the best of our knowledge, it was the first SWE study focusing on individuals with healthy skin. The results above provide initial reference criteria for the Chinese population; in future systemic sclerosis studies, they may be used as references for elastic modulus assessment by SWE. Second, this work was the first study to evaluate potential factors that may affect skin elasticity determined by SWE, including the site, sex, age, BMI, and skin thickness.

The results indicate that the site, sex, and age had significant impacts on the skin elastic modulus measured by SWE, which corroborated findings from appearances on mechanical devices that healthy skin elasticity depends on the site, sex, and age.[7,8,24–29] The elastic modulus values were different at various sites, with the descending order being finger, chest wall, forearm, and abdominal wall. Ewertsen et al[30] showed that the shear wave speed was increased in the soft tissue above the bone; meanwhile, Sun et al[31] reported that skin elastic modulus values were higher with reduced subcutaneous fat thickness. Therefore, the highest elastic modulus value found at the finger might be due to low subcutaneous fat thickness and the proximity to the bone; conversely, the abdominal wall had the lowest elastic modulus value because of higher fat deposition and no bone influence. In addition, collagen and elastin fiber architectures, which mainly determine skin's mechanical properties, may vary at different sites, which might be responsible for the differences in mechanical properties.[32]

Skin elastic modulus values for male participants were significantly higher than those for female participants in all examined sites except for the finger. The results obtained for the dorsal forearm were similar to findings by Luo et al.[19] In that study, the shear modulus (u) values of the normal skin were 4.8 kPa in female participants and 5.2 kPa in male participants, as calculated by the formula $u = \rho c^2$, where ρ is the volume density of the tissue, and c is the propagation speed of the shear wave; these values were approximately one-third of the elastic modulus (E) values in this study, which were derived by the formula $E = 3\rho c^2$. The skin collagen content is lower in women than in men because of sex hormones,[24,28] which may be responsible for the differences in mechanical properties of the dermis layer between men and women.

In the human dermis, cutaneous aging involves intrinsic and extrinsic agents. Cutaneous intrinsic aging is mainly caused by dermal atrophy due to collagen loss, degeneration of the elastic fiber network, and loss of hydration.[33–35] This process might explain why the skin elastic modulus values decreased significantly in individuals older than 50 years compared with the 20- to 50-year group at the finger and forearm. Skin elastic modulus values were different

Figure 5. Scatterplots of skin elastic modulus values and skin thickness in healthy volunteers at different sites.



between the 3- to 19- and 20- to 50-year groups at each site, although only the finger showed a statistically significant difference. This finding might be indicative of the low and gradual process of growth hormone–increased skin elasticity.[36] In the chest and abdominal walls, similar modulus values among the different age groups might have been due to insufficient subdivision. We also analyzed the effects of the BMI and US-measured skin thickness on skin elasticity assessed by SWE but obtained no significant correlation.

In the case of the high-frequency Aixplorer US system application, the shear wavelength (1–a few mm) was larger than or equal to the skin thickness (0.7–2.7 mm). In this case, when shear waves propagate along the skin, strong reflections and mode conversions at the interface generate partial guided wave propagation. These effects are referred to as the Lamb wave propagation effect.[37] Taking into account the above effects on the shear wave speed, Mo et al[38] found that elastic modulus values in thin layer gelatin-agar phantoms measured by SWE decreased with decreasing sample thickness. However, this study showed that the highest elastic modulus value was found at the finger, with the thinnest skin among the assessed sites, whereas the lowest value was observed at the abdominal wall, with the thickest skin. This discrepancy could be related to plate thickness (3–50 versus 0.7–2.7 mm) and probably the intrinsic structure of plates, as well as the type of media surrounding the plates (liquid versus subcutaneous fat). The finding of the lack of a significant correlation between the elastic modulus value and skin thickness was in line with a previous study assessing the effects of the thickness of another human thin-layered soft tissue (bladder neck) on its elastic modulus.[39] In 2004, Gennisson et al[15] also reported that skin elasticity measurements correlated with local and intrinsic parameters that were not influenced by skin thickness. These results suggested that the skin thickness and BMI may not be very important factors influencing skin elasticity. Future studies with larger sample sizes are required to confirm these findings.

In this study, results were reported as the Young modulus, which is expressed in kilopascals, by using the elastic shear wave equation $E = 3\rho c^2$. The equation is suitable for a local area that is considered homogeneous, linear, isotropic, and elastic.[40] Nevertheless, this formula is no longer valid for transverse isotropic and longitudinal anisotropic soft tissues such as muscles and tendons because of a fiber arrangement along the same direction.[41,42] Studies have considered the skin an anisotropic material because of skin tension lines, including Langer lines, based on the parallel arrangement of collagen fibers.[42,43] Meanwhile, Piérard and Lapiere[44] suggested that collagen bundles and elastic fibers were oriented multidirectionally in the normal human skin, without an evident difference in the nature and number of fibers oriented in any direction, even if skin tension lines have anatomic support in the reticular dermis, which consists of a preponderant straightening of fibers and thin bundles. Meanwhile, our preliminary data showed that the differences in the Young modulus between the transverse and longitudinal sections were not statistically significant at most sites, including the forearm, anterior chest, and abdomen.[21] In the study by Luo et al,[19] the differences in normal skin elasticity along directions parallel and perpendicular to the Langer lines in both in the dorsal and volar portions of the forearms were inconsistent between male and female participants. Therefore, whether the Young modulus reflects skin elasticity remains controversial and needs further research. Certainly, only using the Young modulus and not considering the shear wave velocity was one of the limitations of this study.

The skin elastic modulus values obtained by SWE at different sites in our study were consistent with the findings from our previous study.[21] The elastic modulus values measured for the abdomen were similar to findings by Sun et al[31] by applying the same vendor's system (Aixplorer; 9.5 ± 3.1 versus 12.4 ± 2.7 kPa). Statistically significant variability has been observed in quantitative elasticity measurements among the different vendors' elastographic systems.[45,46] Different US manufacturers apply proprietary patented calculation modes and excitation frequencies, and system-dependent biases might exist (such as a speckle bias, hardware fluctuations, phase aberration, pulse repetition frequency errors, and beamforming errors), which might result in these differences.[46–48] On the other hand, the differences in terminology, reported parameters, and shear wave frequency reported in the literature make comparisons of results obtained by different elastographic techniques challenging.[47,49] Thus, it is not reasonable to compare our findings directly with other studies that applied different elastographic techniques or commercial systems, which suggests that the reference range in our study must only be used in an analysis conducted with the same system and same transducer (Aixplorer system with a 15–4-MHz linear transducer).

In recent years, some studies proved the feasibility of strain elastography in the identification of benign and malignant solitary skin lesions, which showed malignant skin lesions with less strain compared with benign lesions.[50–52] Iagnocco et al[17] found that strain elastography could can assess skin involvement in systemic sclerosis only in those areas where the dermis was known to be thicker and where the bone hyper-reflection was minimal (forearm), but it could not be applied to those sharply curved parts and those with high hyper-reflection (finger). In strain elastography, the strain value or strain ratio, but not the absolute elastic modulus, is acquired by applying perpendicular compression from the transducer. The qualitative or semiquantitative elastographic appearance makes it impossible to evaluate the extent of skin fibrosis in systemic sclerosis and the elastic modulus of skin local lesions accurately. The strain ratio is not influenced by the ROI depth, whereas shear wave speeds decrease with increasing scanning depth.[30] Even so, it is not essential to consider the influence of the ROI depth in SWE studies of skin and skin diseases because skin is the most superficial organ in the human body. In summary, skin SWE may have great potential for systemic sclerosis and skin local lesions.

In skin imaging, the shear wave speed in the near field of the transducer may be affected by near-field effects, in which a near-field region that is associated with the wavelength exists in wave propagation. Xu et al[53] demonstrated the near-field effect of a Lamb wave in a 2-layer structures (such as a cartilage-bone structure or skin with underlying bone), in which the wave speed increased sharply to a peak speed and then dropped to the Rayleigh wave speed at a higher wave frequency, although the wave speed increased and approached the Rayleigh wave speed at a higher wave frequency in a single-layer structure (such as arterial wall, myocardium, cartilage, or skin without underlying bone).[53,54] Therefore, in skin SWE, we should not only take into account the Lamb wave propagation effect, as mentioned previously, but also consider the near field of the Lamb wave in certain skin with underlying bone, such as the finger. According to the study by Hao et al,[53] applying a higher loading wave frequency than the critical frequency will be valid to prevent the occurrence of the near-field effect and improve the accuracy of the wave speed measurements.

Other emerging methods available for noninvasively quantifying skin elasticity include mechanical tools and optical coherence elastography. Mechanical tools (torsion or suction test instrument and durometer) are able to provide elastic properties of the entire skin as well as the hypodermis in certain cases by estimating the spatially averaged elasticity but not an individual skin layer (eg, the dermis).[4,15,19] Optical coherence elastography as well as SWE may be used to perform in situ elasticity measurements of the dermis [55,56]; however, as the image acquisition time of optical coherence elastography is relatively long, its routine clinical application would be difficult.[56,57] Future studies are likely to be interesting in the comparison and correlation of results among different skin elasticity measurement techniques to understand which metric characterizes each elastic property.

Although to our knowledge, this work was the first SWE study to focus on assessing the effects of potential factors (site, sex, age, BMI, and skin thickness) on normal skin elasticity in healthy volunteers, it had some other limitations. First, the number of volunteers investigated was small, and the effects of confounding factors such as age might have been underestimated. Inclusion of a larger study population with wider age and BMI ranges would be crucial to validate these results and might alter the significance of such confounding factors on skin elasticity. Second, only 4 sites were examined, which were less than in the modified Rodnan clinical skin scoring system. Future studies of normal reference ranges for the 17 modified Rodnan skin score sites are warranted. Third, the evaluations of the volunteers did not involve skin tests and lacked histologic corroboration.

In conclusion, skin elastic modulus values assessed by SWE in healthy individuals are affected mainly by the skin site, sex, and age, with the descending order being the finger, chest wall, forearm, and abdominal wall. The mean elastic modulus values for normal skin were 30.3, 14.8, 17.8, and 9.5 kPa for the finger, forearm, chest wall, and abdominal wall, respectively. In addition, skin elasticity was higher in men than in women at each site and was elevated in participants aged 20 to 50 years than in the other age groups at the finger.

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